

BFGS Optimization of 2D Chebyshev CP Decomposition with Anisotropy

Documentation for `bfgs_cp_chebyshev_2d_matrixC.py`

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1 Problem Setup

We consider the approximation of the anisotropic test function

$$f(x, y; C) = \frac{1}{1 + \|C \cdot [x, y]^\top\|^2}, \quad (1)$$

where $C \in \mathbb{R}^{2 \times 2}$ is a matrix encoding anisotropy (e.g. scaling, stretching, or rotation). Different choices of C control the shape of the level sets of f .

Examples of C matrices

The script evaluates the method for several representative cases:

$$\begin{aligned} C_{\text{vertical}} &= \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}, & C_{\text{horizontal}} &= \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}, \\ C_{\text{tilted}} &= \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} R(30^\circ), & C_{\text{steep}} &= 5I, & C_{\text{flat}} &= 0.2I, \end{aligned}$$

where $R(\theta)$ is a standard 2×2 rotation matrix.

2 Chebyshev Basis

For a univariate variable $x \in [-1, 1]$, the degree- d Chebyshev Vandermonde matrix is defined as

$$T(x) = [T_0(x) \ T_1(x) \ \cdots \ T_d(x)], \quad (2)$$

where T_k denotes the k th Chebyshev polynomial. The script constructs such Vandermonde matrices for both x and y directions.

3 CP Decomposition Ansatz

We seek an approximation of f using a CP (CANDECOMP/PARAFAC) decomposition:

$$\hat{F}(x_i, y_j) \approx \sum_{r=1}^R \lambda_r \left(\sum_{p=0}^{d_x} A_{p,r} T_p(x_i) \right) \left(\sum_{q=0}^{d_y} B_{q,r} T_q(y_j) \right), \quad (3)$$

where:

- R is the CP rank,
- $A \in \mathbb{R}^{(d_x+1) \times R}$ and $B \in \mathbb{R}^{(d_y+1) \times R}$ are coefficient factors,
- $\lambda \in \mathbb{R}^R$ are optional scaling weights.

4 Optimization with BFGS

We define the mean squared error (MSE) on the training grid:

$$\mathcal{L}(A, B, \lambda) = \frac{1}{NM} \sum_{i=1}^N \sum_{j=1}^M \left(f(x_i, y_j; C) - \hat{F}(x_i, y_j) \right)^2. \quad (4)$$

The script minimizes $\mathcal{L}$ using the BFGS quasi-Newton method (`scipy.optimize.minimize` with `method="BFGS"`).

5 Reference Solution

For comparison, a direct Chebyshev interpolation is computed by solving a least-squares problem:

$$\min_{\alpha} \|A\alpha - f\|_2, \quad (5)$$

where A is the Kronecker product of Chebyshev Vandermonde matrices in x and y . The resulting coefficient tensor serves as a reference “ground truth”.

6 Evaluation Metrics

The script computes several metrics:

- Training RMSE on the interpolation grid,
- Test RMSE and max error on independent points,
- Dense grid RMSE,
- L^2 coefficient error:

$$\|C_{\text{interp}} - C_{\text{recon}}\|_2,$$

and relative error with respect to $\|C_{\text{interp}}\|_2$.

7 Outputs

For each run, the following are saved in a timestamped directory:

- `metrics.json` containing all error values,
- `coeffs.json` containing CP factors and reconstructed tensor,
- `config.json` with experiment settings,
- A plot (`reconstruction.png`) showing:
 1. Left: the original function $f(x, y)$,
 2. Right: the reconstructed CP approximation.

8 Example Usage

The script loops over the chosen C matrices. For each case:

```
python bfgs_cp_chebyshev_2d_matrixC.py
```

A timestamped folder is created under `bfgs_results/`, containing all JSON data and plots.

9 Remarks

- Setting $N = M = 64$ with $d_x = d_y = 63$ corresponds to Chebyshev interpolation of degree 64 in each dimension.
- CP rank $R = 10$ controls the approximation complexity.
- The method illustrates how tensor decompositions combined with Chebyshev bases can efficiently approximate anisotropic functions.